Viva Questions and Answers

IT 207 Human Intelligence

1. How does the system connect human intelligence to AI?

It uses selective-attention ideas for ROI-based lane perception, working-memory ideas for recent traffic history, learning for forecast model code, and decision-making for weighted signal allocation. These are computational analogies, not measurements of human intelligence.

2. What was already present in the workspace?

Detection, tracking, analysis, prediction, optimization, SUMO hooks, API, dashboard, and YAML configuration were present. The README, tests, model weights, data, and submission documents were absent.

3. Which detector and tracker does the project configure?

YOLO11n with COCO vehicle classes is configured; the tracking module uses Ultralytics ByteTrack. No weights or labeled video were supplied, so neither was evaluated.

4. What do the queue and speed values mean?

They are proxies. Queue length assumes five metres per vehicle and speed is a count-based heuristic unless calibrated positional data exist. They are not physical measurements.

5. What controllers were compared?

Fixed-time, rule-based, and AI-assisted weighted allocation. All used the same arrival rates and random seed per scenario.

6. What are the main experiment results?

AI-assisted performed better than fixed-time on delay and throughput in the uneven-demand scenario, but worse on both in medium and heavy balanced scenarios. One seeded run is preliminary and does not establish general superiority.

7. Why are there no mAP or IDF1 scores?

There was no video, ground-truth annotation, or YOLO weight file in the workspace, so calculating those metrics would be unsupported.

8. Why was SUMO not used for reported results?

SUMO and TraCI were not installed, and the generated network did not have a verified traffic-signal phase plan. The results therefore come from the software queue simulator.

9. What is the key limitation of the controller evaluation?

The simulator simplifies arrivals, discharge, and phases; it omits turning conflicts, pedestrians, startup loss, spillback, calibration, and uncertainty across repeated seeds.

10. What ethical concerns apply to traffic video?

Footage can expose faces, plates, and movement patterns. Minimize retention, restrict access, document lawful purpose, and measure performance across camera, lighting, class, and traffic conditions.

11. Does the system control real traffic lights?

No. It is software-only and returns recommendations; there is no hardware or physical-controller integration.

12. What would you do next?

Install the supported Python 3.11 dependencies, add licensed annotated video, verify model paths, measure detection and tracking metrics, train forecasting on real time-ordered data, and run repeated experiments in a validated SUMO network.